

Tables and Temp Tables

(Recaps database structure) (Recap DDL)
(Recaps levels of database architecture)

TABLE

structured collection of data,
similar to spreadsheets, grid.

DATABASE FILES

Database data is actually a massive set of files containing all the information about the database.

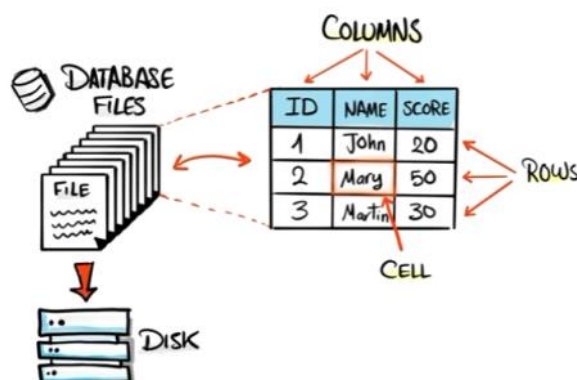
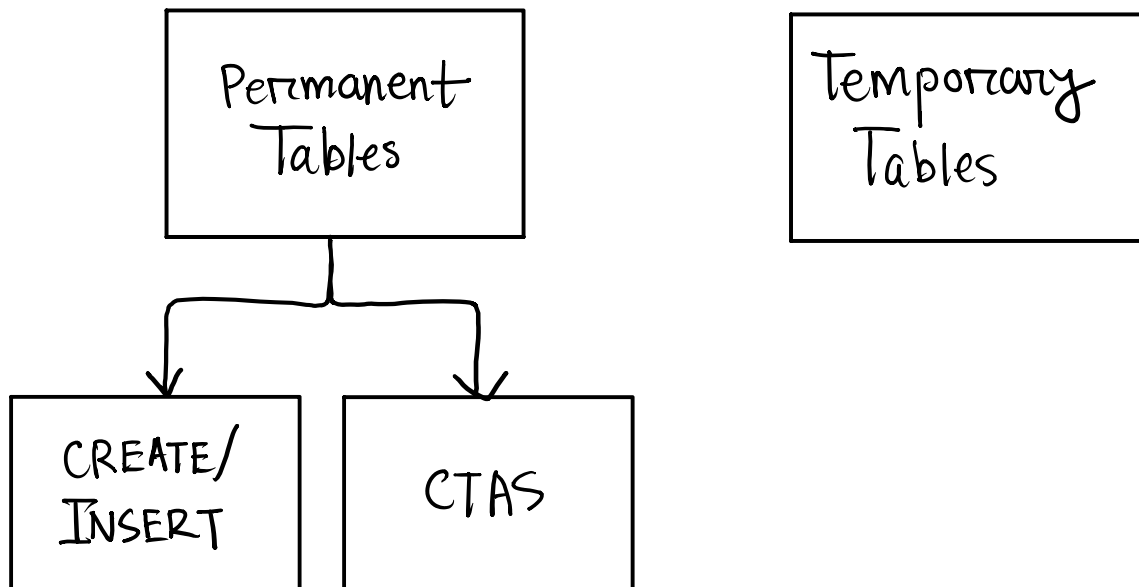


TABLE TYPES



CREATE/INSERT

USE create, insert keywords.

CTAS

[Create Table As SELECT]

Create a new table based on the result
of a query

CTAS vs Views

Views:

View query is executed every time the view is called in a user query.

→ table is temporary. Destroyed after use.

→ Slow. But flexible. Table created at run-time.

CTAS:

This generates a physical table. That table can be directly queried. No need to re-gen the table as the table is already a physical table.

→ Takes up storage though; the table.

→ Fast. But rigid. Table is physical.

Syntax:

Create / Insert

DDL { CREATE TABLE table-name (

<var-name> <data-type> <NULL> ,

=====

)

DML { INSERT INTO <table-name>

VALUES (____, ____)

CTAS

Works on

MySQL, Postgres, Oracle

CREATE TABLE <table-name> AS

(

SELECT _____

FROM _____

WHERE _____

)

NOT in SQL Server



looks very familiar.
(just like VIEW, But table.)

in SQL Server,

SELECT _____

INTO <table-name>

without the AS ,

specify table columns,

then use INSERT to

populate this table.

INTO <table-name>

FROM _____

WHERE _____

populate this table.

This queries another table. The result becomes another table.

Use cases:

1. Performance optimization.

Views can become slow.

For that we can use CTAS.

Better use a combination of both, where possible.

Can find a list of valid object types for object-id here.

[sys.objects \(Transact-SQL\) - SQL Server | Microsoft Learn](#)

For view → 'V'

For table → 'U'

(user defined)

Refresh CTAS

TSQL

Need to be done every time
main user define tables are updated.

Use case

2. Create a table snapshot.

Create a snapshot and work on it.
Let the main table update as much
as it wants.

3. Create Physical data marts.